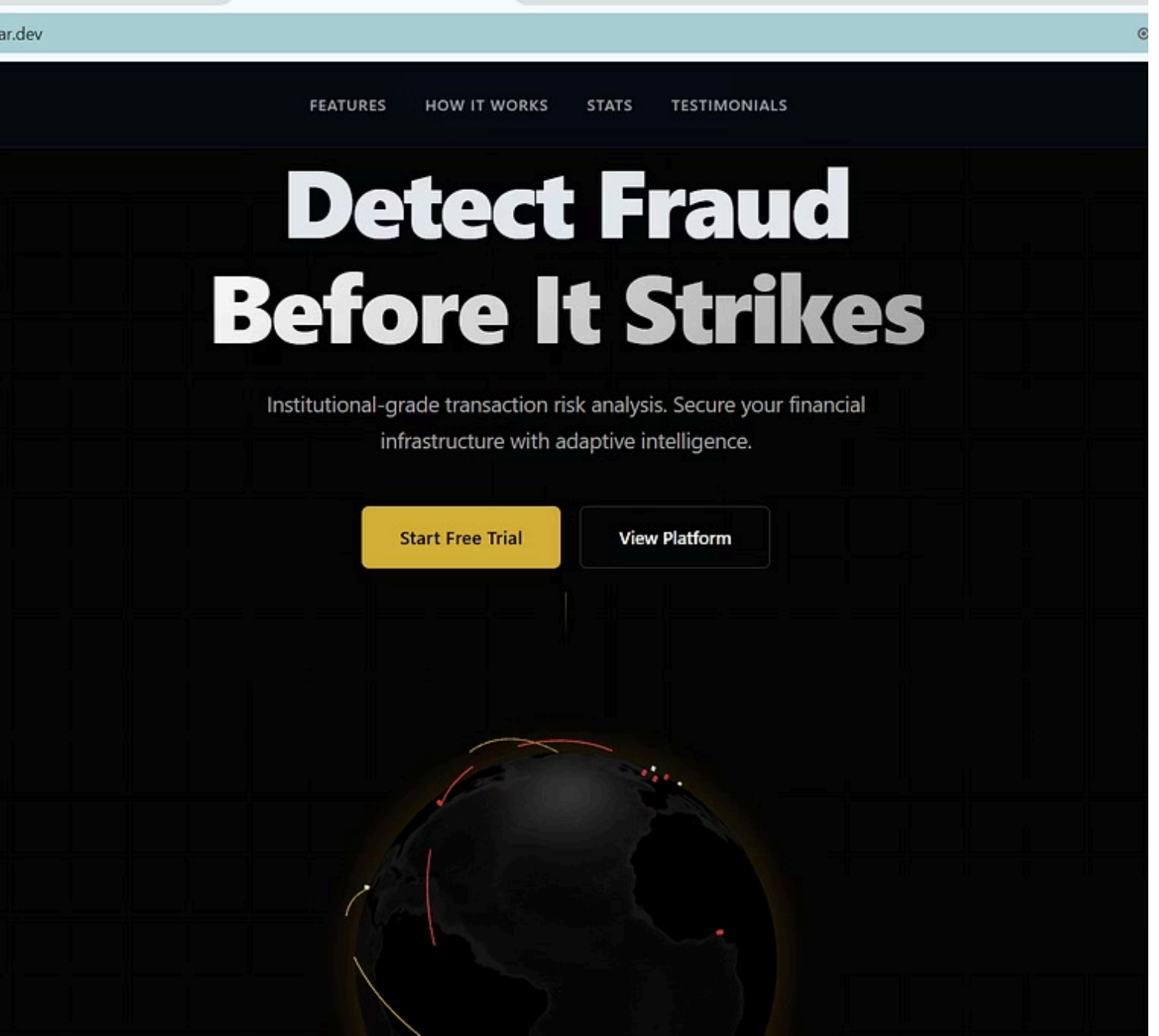

FraudGuard AI

AI-Powered Fraud Detection Web Application

B.Tech CSE

INT - 428

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The Problem: Fraud Is Escalating Fast

Rising Online Fraud

Global digital fraud losses exceeded **$48 billion** in 2025, growing at an alarming rate year over year.

Real-Time Gap

Traditional rule-based systems fail to detect novel fraud patterns in real time, leaving transactions vulnerable.

Need for Intelligence

Manual review is slow and unscalable. Intelligent, automated fraud detection is no longer optional — it's essential.

Introducing FraudGuard AI

FraudGuard AI is a full-stack web application that leverages large language models and modern UI design to detect fraudulent transactions in real time — combining analysis, visualization, and conversational AI in one platform.

Real-Time Detection

AI analyzes transactions instantly, returning a fraud verdict, risk score, and confidence level.

Interactive Dashboard

Visual charts and risk indicators give users a live overview of transaction health.

AI Chatbot + Voice

An integrated chatbot powered by Llama, with Whisper-based speech-to-text for hands-free input.

Key Features

Transaction Analyzer

AI-driven engine that classifies each transaction as Fraud or Safe with a risk score and confidence percentage.

Interactive Dashboard

Dynamic Recharts-powered visuals displaying transaction history, fraud trends, and risk distribution.

AI Chatbot

Conversational assistant that answers fraud-related queries and guides users through the platform.

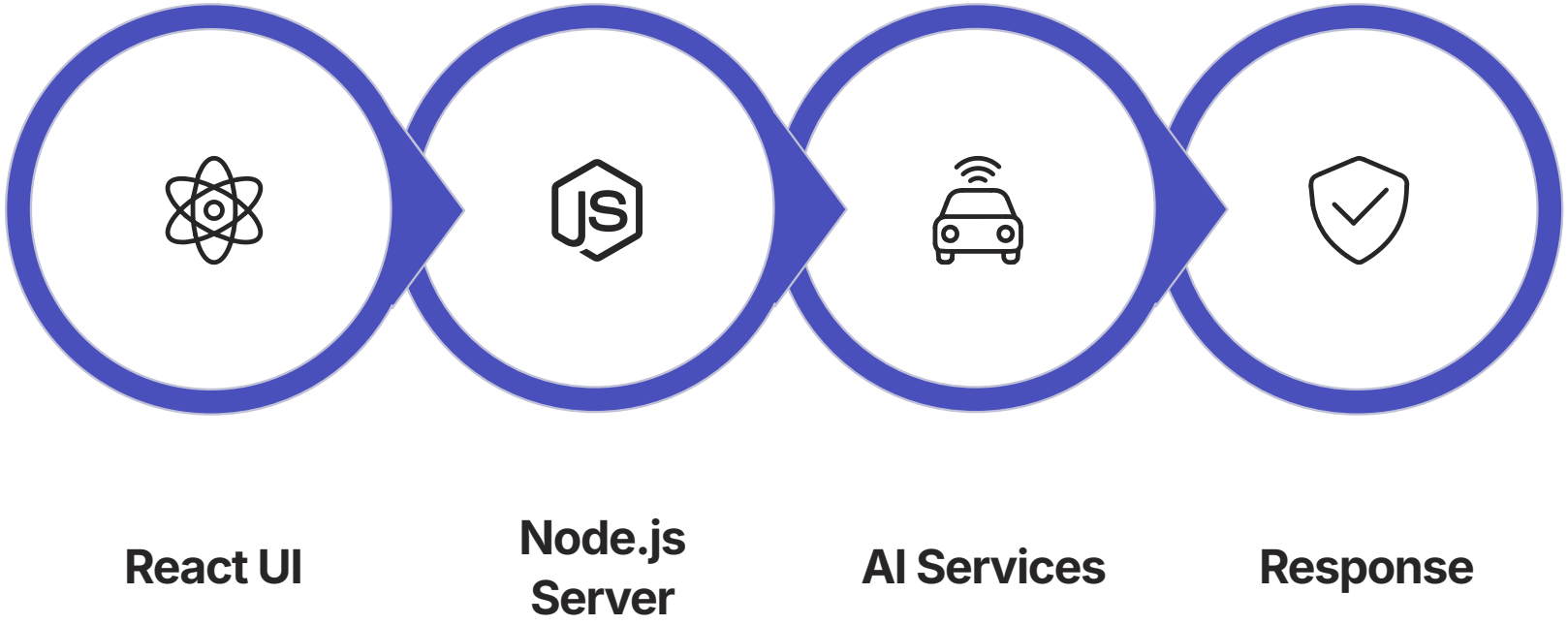
Voice Input

Whisper-powered speech-to-text enables users to input transactions hands-free via voice commands.

Global Fraud Globe

3D globe visualization highlighting geographic hotspots of detected fraudulent activity worldwide.

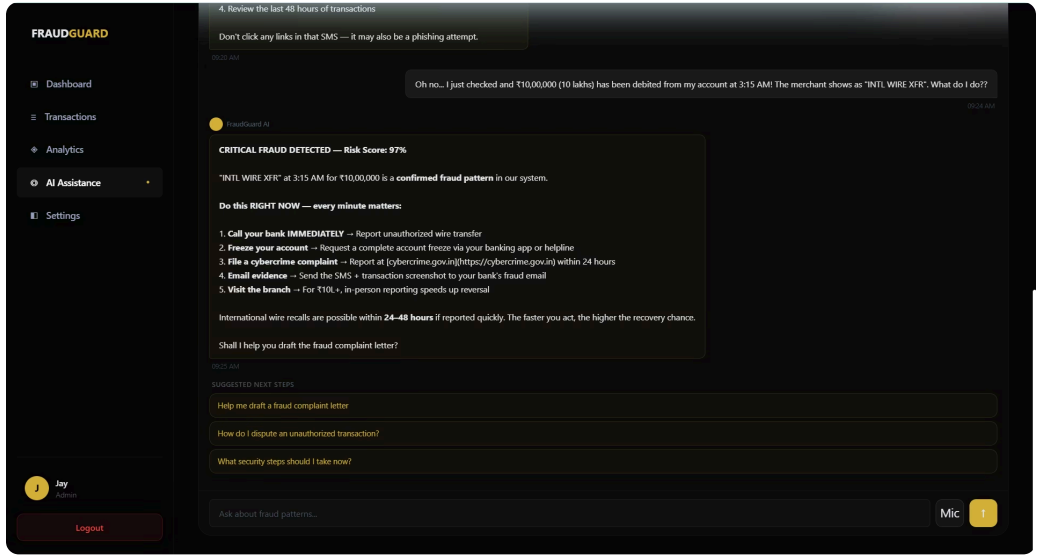
System Architecture



The architecture follows a clean separation of concerns — the React frontend handles all UI interactions, while the Node.js/Express backend orchestrates communication between the client and AI inference APIs, ensuring modularity and scalability.

Tech Stack

A modern, production-grade stack chosen for performance, developer experience, and AI integration capabilities.



Frontend

React, Vite, Recharts, Framer Motion

Backend

Node.js, Express.js REST API

AI Models

Groq (analysis + chat), e)

Deployment

Vercel — serverless, zero-config hosting

How It Works

User Input

User enters transaction details — amount, merchant, location — via form or voice.

API Request

Frontend sends a POST request to the Node.js backend endpoint for analysis.

AI Analysis

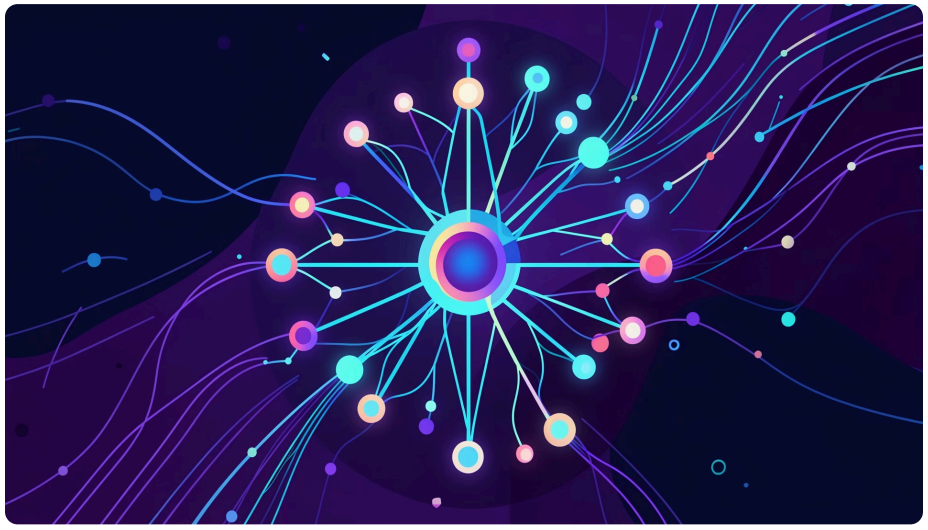
Groq processes the transaction and returns a fraud verdict, risk score, and confidence level.

Result Display

Dashboard renders the verdict with animated risk indicators and updates the transaction history chart.

AI Integration Deep Dive

FraudGuard AI uses two specialized AI models, each selected for its unique capability in the fraud detection pipeline.



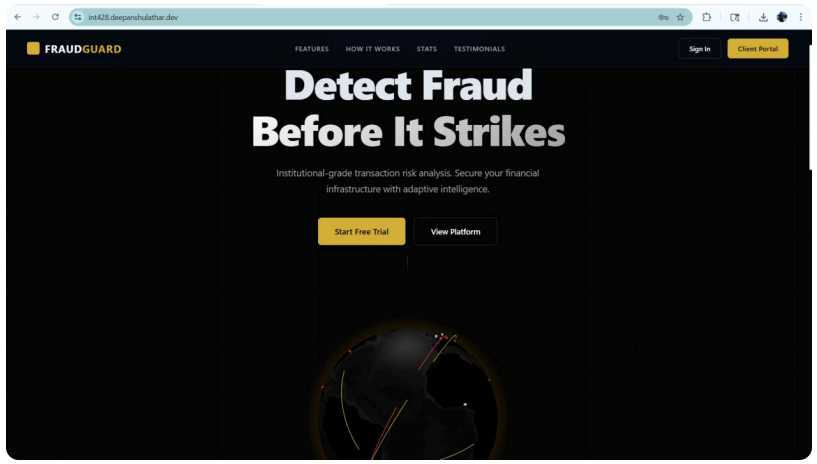
Groq — Fraud Analyzer

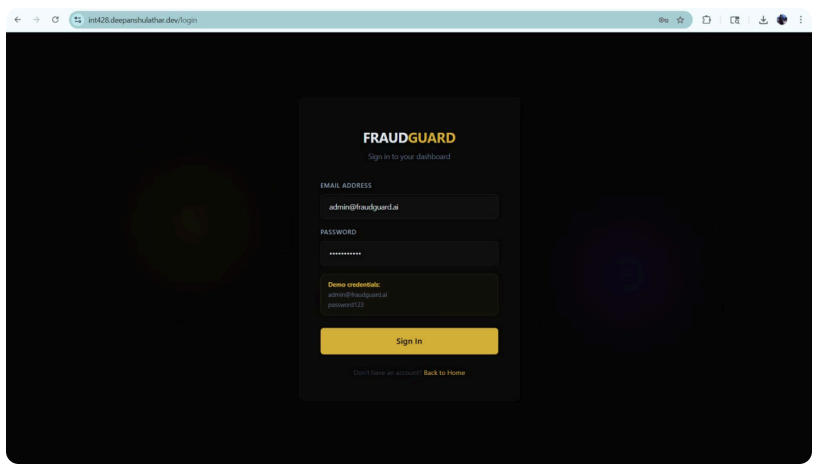
Runs at **temperature = 0** for fully deterministic, reproducible fraud verdicts. No hallucinations — just consistent analysis.

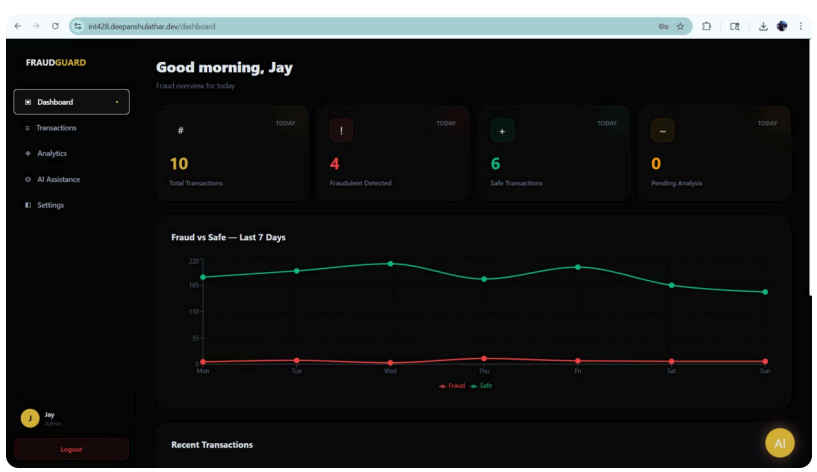
Groq — Chatbot

Runs at higher temperature for natural, conversational responses to user queries about fraud and platform usage.

UI/UX Highlights & Demo Flow







Demo Walkthrough

01

Login

User authenticates and lands on the main dashboard.

02

Analyze Transaction

Enter or speak transaction details; submit for AI analysis.

03

View Result

Dashboard displays verdict, risk score, and animated confidence ring.

04

Chatbot Interaction

Ask the AI assistant follow-up questions about the flagged transaction.

Glassmorphism design with Framer Motion animations delivers a polished, fintech-grade experience.

Strengths, Limitations & Future Roadmap

Strengths

Full-stack AI implementation · Real-time analysis · Modern glassmorphism UI

Limitations

Hardcoded auth · In-memory storage · Static data · Basic security model

Future Improvements

- **JWT Authentication** — Replace hardcoded login with secure, token-based auth
- **PostgreSQL Database** — Persist transactions and user data for longitudinal analytics
- **Real Data Analytics** — Replace static charts with live, database-driven visualizations
- **Mobile Responsiveness** — Extend the glassmorphism UI to mobile and tablet viewports
- **Enhanced Security** — Rate limiting, input validation, HTTPS enforcement, and audit logs

FraudGuard AI is architected for growth — every limitation has a clear, implementable path forward toward a production-ready fintech platform.